

Doom PWAD QA Report

Run 42614da5-07ac-4b4c-a7c6-9b1ee9aefda9 | Map MAP01 | Outcome stuck

Executive Summary

This is an autonomous QA test of MAP01 using a lockstep LLM/MCP test harness. The test was designed to evaluate map navigation, combat engagement, resource balance, and secret discovery within the provided environment, resulting in a 'stuck' outcome after 268 seconds of runtime.

Pass/fail:

map_navigation: FAIL

combat_engagement: FAIL

resource_balance: FAIL

secret_coverage: FAIL

overall_verdict: FAIL

- The run ended with outcome 'stuck' after visiting 5 coarse position cluster(s).
- 0 kill(s) recorded against 13 enemy/enemies that spawn at difficulty 1.
- Selected-difficulty health_ratio=0.0844, ammo_ratio=0.1200.
- 0 secret(s) found.

No technical problems encountered during this test run.

Test Item And Environment

The test covered MAP01 using the Freedoom2 IWAD on difficulty 1. The map features 14 enemies, 12 items, 1 key, and 1 secret sector, spanning 1536x872 units with a complexity of 45 sectors and 174 linedefs.

The environment utilized ViZDoom, the gemini-3.1-flash-lite model, and an MCP tool interface. The test was conducted at difficulty level 1 with a 500-tick cap.

Metric	Value	Metric	Value
Status	completed	Duration	268
Actions	41	LLM calls	41
Final HP	101	Kills	0
Position samples	115	Movement units	598.0
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/42614da5-07ac-4b4c-a7c6-9b1ee9aefda9.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	14	13	1 hidden by skill/multiplayer flags
Items	12	12	0 hidden by skill/multiplayer flags
Secrets	1	1	Sector special 9
Geometry	174 linedefs	45 sectors	1536 x 872 units

Selected-difficulty estimate: hard. Health ratio: 0.0844. Ammo ratio: 0.12.

Coverage Evaluation

Coverage was extremely limited, confined strictly to the starting sector. No enemies were engaged, and no secrets were discovered, as the automated playthrough did not trigger any functional doors or exits to reach the rest of the map. The coverage is insufficient for a full map quality assessment.

Objectives Covered

- **Document resource balance issues:**

Objectives Omitted

- **Navigate the entire map:** Navigation failed due to a critical progression blocker at the start sector.
- **Engage all enemy types encountered:** Enemies were unreachable from the starting sector.
- **Find secrets where accessible:** Secrets were not reached due to navigation failure.
- **Stress-test geometry, doors, lifts, and switches:** Doors were found to be non-functional, causing the navigation failure.

Uncovered attributes: Map-specific scripting and geometry connectivity could not be tested beyond the starting room. The observation of non-functional doors indicates a significant structural design issue.

Defects And Risks

A total of 5 defects were detected, including 2 critical-severity progression blockers and 3 minor/trivial issues. The most impactful defect is the unopenable starting area door, which prevents all map progression.

Patterns: The defects are primarily localized to the starting area, specifically involving collision issues and non-functional door sectors that result in a hard progression stall.

Severity	Priority	Type	Tick	Description
1	1	softlock_navigation	41	Run stalled after repeated stuck decisions The run produced repeated stuck events with no meaningful progress. This usually indicates blocked navigation, unreachable objectives, or an automated control loop that should be reviewed with the recording and action trace.
2	2	agent_observed_geometry	129	Automated playthrough observed geometry issue [GEOMETRY] Starting area confinement. At tick 106, position (50.47, -17.12): Player is blocked by geometry/collision while attempting to reach visible door paths, preventing progression. Player impact: critical. Severity: critical.
2	2	agent_observed_progression	148	Automated playthrough observed progression issue [PROGRESSION] Starting area door non-functional. At tick 147, position (-76.35, -223.71): The initial sector has no reachable path to the main map; none of the local doors respond to USE commands. Player impact: Cannot leave starting area. Severity: critical.
3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 1 of 14 enemies from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
3	3	unreachable_secret	173	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Starting sector exits:** high Critical progression blocker prevents completion of the map.

Activity Log Summary

1) Static analysis phase (Tick 0), 2) Map exploration phase (Ticks 15-173), 3) Combat phase (None), 4) Defect detection phase (Concurrent), 5) Report generation phase.

Exploration was limited to the starting sector throughout the entire duration due to the progression block.

Tick	Event	Position	HP	MCP Action	Stop reason
39	item_pickup	-58.1, -68.3	101	move_to	arrived
41	stuck	-58.1, -68.3	101	move_to	arrived
43	stuck	-58.1, -68.3	101	move_to	arrived
45	stuck	-58.1, -68.3	101	move_to	arrived
47	stuck	-58.1, -68.3	101	move_to	arrived
85	stuck	12.6, -59.7	101	explore	item_found
87	stuck	34.8, -47.9	101	take_action	tics_complete
103	stuck	50.5, -17.1	101	explore	item_found
105	stuck	50.5, -17.1	101	explore	item_found
157	stuck	-80.0, -223.9	101	take_action	tics_complete
173	stuck	-78.0, -223.9	101	explore	item_found

#	Ticks	Tool	Stop reason	Reasoning summary
0	14 -> 15	explore	item_found	Starting exploration of the initial room to gather health and ammo bonuses before engaging the distant hitscan threats.
1	16 -> 24	move_to	arrived	Initial spawn area contains health and armor bonuses which are critical for survival against the high hitscanner count. Moving to secure the HealthBonus at id 2.
2	25 -> 26	move_to	arrived	Securing the nearby HealthBonus to improve survivability before engaging the distant hitscan threats detected in the area.
3	27 -> 28	move_to	arrived	Securing the nearby HealthBonus to build a safety buffer before navigating toward the hitscanner-heavy areas.
4	29 -> 30	move_to	arrived	Securing the nearby HealthBonus to establish a buffer against the high hitscanner concentration before moving further into the map.
5	31 -> 32	move_to	arrived	Securing local resource (HealthBonus) before advancing to engage the hitscanner-heavy areas to the south.
6	33 -> 39	move_to	arrived	The player is gathering nearby HealthBonuses to mitigate the high hitscanner percentage risk before proceeding. Now moving toward the reachable Clip pickup at id 5 to optimize ammo before engaging enemies.
7	40 -> 41	move_to	arrived	Gathering nearby ammo pickup to ensure resource readiness before navigating toward the southern hitscanner-heavy engagement zones.

8	42 -> 43	move_to	arrived	Gathering the nearby Clip (id 5) to ensure ammunition reserves before navigating toward the southern doorways where the hitscanner threats are concentrated.
9	44 -> 45	move_to	arrived	Gathering the nearby Clip (id 5) to maximize ammunition before engaging the primary encounter zone, as per the briefing warning about hitscanner concentration.
10	46 -> 47	move_to	arrived	Gathering the nearby Clip to ensure ammunition reserves before navigating toward the southern doorways where the hitscanner threats are concentrated.
11	48 -> 49	take_action	tics_complete	The player is currently stuck on local geometry while attempting to collect the nearby clip. Moving to a slightly offset position to clear the collision and continue towards the southern combat area.
12	50 -> 51	take_action	tics_complete	The player is currently obstructed by geometry near the start point. Executing a small movement probe to bypass the collision before exploring toward the southern doorways where the hitscanner threats are located.
13	52 -> 54	take_action	tics_complete	The player is currently obstructed by geometry near the starting area. Executing a small movement rotation and forward probe to clear the collision and progress toward the nearby doors and identified threats.
14	55 -> 75	move_to	arrived	Collecting the nearby shell pickup to bolster resources before engaging the hitscanner-heavy area to the south. Rotating and moving to clear the initial collision geometry near the start point.

15	76 -> 77	explore	item_found	The player is currently clear of the start position and needs to begin exploration to locate the keycard and engage threats; heading towards the unexplored eastern section.
16	78 -> 79	explore	item_found	The player is clear of the initial spawn geometry and is now moving to engage the identified hitscanner threats to the south while exploring for the blue keycard.
17	80 -> 81	explore	item_found	The player has finished collecting the nearby shells and is now moving towards the southern area to engage the identified hitscanner threats and explore the path to the blue keycard.
18	82 -> 83	explore	item_found	The player has successfully collected the initial shell pickup and is now moving towards the southern area to engage the identified shotgunners and explore the corridor leading to the blue key.
19	84 -> 85	explore	item_found	Collected initial shells and positioning to engage potential hitscanner threats in the corridor; moving to secure resources and clear the area.
20	86 -> 87	take_action	tics_complete	Progress has not changed across repeated lockstep decisions, so I am forcing a bounded QA recovery probe. A nearby door-like sector is known, so I am pressing USE and nudging forward.
21	88 -> 95	explore	item_found	Navigation is constrained by the geometry. Initiating an exploration move to break the current movement stall and identify paths through the identified closed doors.

22	96 -> 97	explore	item_found	The player is positioned in an area with restricted visibility and nearby closed doors. Using an exploration probe to move towards the south and attempt to trigger door activation or reveal new pathways.
23	98 -> 99	explore	item_found	Current position is cramped with no line-of-sight to the threats identified in the briefing. I will perform a short exploration to navigate around the pillars and attempt to find a line of sight or an accessible door.
24	100 -> 101	explore	item_found	I am performing a controlled exploration to navigate around the immediate central pillars and approach the nearby doors, as there are no visible targets to engage at this current position.

Limitations And Recommendations

All locked doors and teleporters were unreachable due to the initial progression block.

Good Quality Areas

- **Starting area item placement:** Resources are correctly identified and collected by the automated playthrough, confirming basic item interaction functionality.